ENVIRONMENTAL POLITICAL ECONOMY: A RESEARCH AGENDA

Matthew Alampay Davis

1 Climate change and global income inequality

Mitigating anthropogenic climate change and constraining economic inequality are widely recognized as two of the most pressing social priorities of the 21st century. For example, they are foundational to the 2030 Agenda adopted by all United Nations members whose Sustainable Development Goals explicitly resolve to "take urgent action to combat climate change and its impacts" and "reduce inequality within and among countries". These are also understood to be intimately related objectives: the Intergovernmental Panel on Climate Change identifies "widening opportunities between and within countries" as first-order considerations in any sufficient policy response to climate change. Conversely, the World Inequality Report regards climate change as "largely a distributional issue, not only between countries but also within them".

Despite this recognition, the economic analysis of climate change remains remarkably agnostic to distributional effects. To the limited extent that these have been estimated, empirical studies have been typically characterized by a stark tradeoff between subnational disaggregation and global representativeness. Even though global income inequality is now understood to be dominantly driven by within-country disparities, little is known about the extent to which climate change poses differentiated risks to the world's eight billion people beyond what is explained by less than 200 national borders.

This project seeks to initiate an analysis of the inequality implications of climate change and its policy implications. The first stage extends my job market paper which estimates the impacts of climate change on global inequality at subnational levels. The second stage builds on these results, drawing from advances in the climate impacts, optimal taxation, and directed technical change literatures to investigate the implications of income inequality for the optimal design and mix of climate policies.

1.1 Global distributional impacts of climate change

My job market paper makes significant progress towards the distributional analysis of climate change by applying methodological advances in time series methods to income distributional data synthesizing information from survey, national accounts, and tax record sources. These newer methods not only more credibly identify environmental shocks and their dynamic effects on social welfare but also avoid sources of extreme instability known to produce estimates which diverge by orders of magnitude depending on the analyst's specification of the underlying climate function. Taking these methodological refinements to a new setting, my analysis provides the first rigorous characterization of the persistent and regressive global impact of climate change, which is accomplished with minimal parametric baggage.

Figure 1a summarizes the main within-country results from this analysis, visually depicting point estimates of the state-dependent dynamic response of country-percentile income to a 1°C temperature shock for three representative climates. These dynamic inci-

dence curves suggest that individuals in cooler countries are expected to be largely insulated from such shocks regardless of their relative income level. In contrast, individuals in 28° climates, approximately the 80th percentile of national climates in our data, are expected to experience income losses which persist for at least three years. This incidence pattern is highly unequal with the most severe costs concentrating on the poorest deciles within these countries as well as, perhaps surprisingly, a similarly sharp negative effect on the top 1%.

To understand the implications of this incidence pattern for climate change, we need to account for geographic variation in climates and temperature shocks, the spatial incidence of climate change, and the global distribution of income. The other next two panels present the results of such an analysis comparing the trajectory of global income as historically observed between 1981 and 2016 (depicted as black lines) to a counterfactual where greenhouse gas emissions are held fixed at 1981 levels (depicted as colored bands capturing bootstrap uncertainty in the dose-response function). Figure 1b compares a version of Branko Milanovic's famous "elephant curve" to the counterfactual long-term growth. The results strongly suggest that anthropogenic warming over this 36-year period has dramatically deprived the poorest global percentiles of income-earning opportunities that would have raised average incomes by a factor of between 2-3. Less dramatic proportional losses are also estimated for the global 60th-80th percentiles. On the other hand, the global middle class and the very richest percentiles are more likely to have benefited from warming over this period since these income groups are dominantly comprised of inhabitants of cool climates. Figure 1c summarizes the implications of this global incidence pattern for inequality: regardless of the chosen measure of inequality (larger values correspond to greater inequality), results are strongly suggestive that global inequality has already been greatly exacerbated by climate change.

I would use the beginning of my fellowship to further develop and refine these results, firstly by exploring competing theories for the underlying mechanisms behind these inequality effects. For example, the observed negative effect on the top 1% in warmer climates may be explained by a disproportionate capital share of agricultural income or the relative underdevelopment of financial products in developing settings. The historical counterfactuals presented here could also be complemented by a standard projection exercise to estimate the inequality effects that would be realized under different mitigation and adaptation scenarios; this could then be quantitatively expressed as a type of distributional social cost of carbon. From my own publication experience and early feedback on my job market paper, I expect that a refined version of the existing analysis would be a strong candidate for submission to top scientific journals.

1.2 Income inequality and optimal climate policy

There is rapidly growing interest in understanding how economic inequality may impact the optimal policy response to climate change. For example, the United States government in 2023 revised its guidance on the conduct of all regulatory analyses for the first time in 20 years to explicitly incorporate equity weighting which places higher priority on low-income individuals. In the wake of this revision, recent work has shown that incorporating inequality considerations in this manner has dramatic implications for climate policy, increasing the corresponding social cost of carbon by as much as a factor of eight (Prest et al. Science

2024). Thus, there exists a critically underexplored relationship between the regressivity of climate change and the optimal policy response to climate change.

Of particular interest is an undertheorized and overlooked complementarity between redistributional policy—such as fiscal tax and transfer schemes—and the efficacy of a given carbon price: if equity weighting implies a higher social cost of carbon, the underappreciated imiplied corollary is that redistributive policy would reduce the efficient carbon tax.

This recalls similar complementaries theorized in the recent literature on directed technical change (Acemoglu et al. AER 2012, Aghion et al. JPE 2016, Acemoglu et al. JPE 2016) wherein green subsidies are modeled to raise the elasticity of substitution between fossil fuels and renewable energy sources so that the same degree of clean-energy substitution can be induced with a significantly smaller carbon price. In the same manner, this analysis would explore the possibility that the efficient means of internalizing the climate externality involves a coordinated program of compensatory policy in conjunction with standard carbon pricing instruments. This would be tremendously important as a means of overcoming the historical political opposition to the administration of carbon taxes on the grounds of its potentially regressive incidence. In additional, by relying on unilateral within-country policy, the optimal policy mix would be independent of slow-moving and consensus-based intergovernmental action and thus more realistically implemented in the immediate time scale required for averting the worst climate damages.

2 Political instability under environmental stress

My second dissertation chapter builds on recent political economy work by Brückner and Ciccone (ECMA 2011) and Acemoglu et al. (JPE 2019), which observes that democratizations tend to be preceded by negative transitory shocks. A compelling theoretical explanation for this phenomenon refers to a "democratic window of opportunity" wherein the shock decreases the opportunity cost of contesting political power; the worst impacted have nothing to lose, so to speak.

This naturally fits within the distributional program of climate research I seek to build upon. Indeed, anecdotes abound of environmental disasters inducing grassroots political activity such as the droughts in southern India which have politically mobilized farmers to march and camp outside New Delhi demanding a governmental response or the extended drought in Syria which is often identified as one catalyst for the Arab Spring. This project seeks to investigate whether the demand for democracy is a generalizable feature of environmental shocks and its implications for future trends in the global climate.

degree to which environmental shocks impact the probability of transitions in or out of democracy and whether these impacts are driven by the distributional incidence of these stresses. Incorporating our empirical results into a hidden Markov model, we consider their implications in the context of climate change and rising global populism.

2.1 Environmental shocks and political regime change

In an exploratory analysis, the democracy panel datasets from the Polity IV Project; Cheibub, Gandhi, and Vreeland (2010); and Acemoglu et al. (2019) are used to identify discrete country-year democratization events and democratic reversals which are then regressed

against temperature shocks as constructed in my job market paper. The local projection regression results depicted in Figure 2 indeed recover a significant impulse response of the propensity for autocracies to democratize (panel 2a) and democracies to backslide (panel 2b). Interestingly, the signs of these effects appear to be highly state-dependent such that a hot shock has opposite effects on the probability of democratic transition for hot and cold countries. The method of estimating persistence of these temperature effects introduced in my job market paper also demonstrate the transitory nature of these effects, consistent with the aforementioned "window of opportunity" theory. Analogous results for year-to-year changes in continuous democracy "scores" recover similar effects.

2.2 The demand for democracy

A natural question that arises is one of underlying mechanisms. For example, are these effects driven by the 'demand' for democracy among a constituency or the 'supply' from governing elites, perhaps as a concession to consolidate power amidst turmoil? To explore the demand channel, I draw from recent work (Acemoglu et al., REStud 2024) which uses data from the Integrated Values Survey dataset to quantiatively represent national sentiments towards democracy, authoritarianism, and democratic inclusiveness. I intend to combine this with data from the Varieties of Democracy dataset to determine whether the same environmental shocks induce consistent and quantifiable changes in national sentiments towards democracy, authoritarianism, and democratic inclusiveness which presage democratic change. I intend to also use the inequality results from my job market paper to investigate the degree to which distributional impacts of these shocks mediate these effects.

2.3 Climate change and democracy

Depending on the results of these analyses, empirical results which deliver temperature-induced dynamic transition probabilities in and out of democracy may then be integrated with projections of future mitigation and adaptation scenarios. For example, a simple hidden Markov model could accommodate these transition probabilities to simulate the likely impact of climate change on the global spread or 'steady state' of democracy in the 21st century.

3 Environmental disruption of gender norms in the Global South

This project, joint with political scientist Tanushree Goyal at Princeton, locates itself in a large and active research area intersecting the economics of gender and development. Recognizing that traditional indicators of economic development correlate with conventional measures of gender equality, it is widely understood that economic growth alone fails to provide a satisfying explanation for the transmission or evolution of gendered norms.

Perhaps nowhere is this more stark than in South Asia where cultural conservativism regarding gender roles has displayed remarkable persistence despite rapid economic gains over recent decades. Indeed, compared to other countries with comparable levels of economic

development, South Asian countries lag far behind in terms of female labor force participation, social integration, women's empowerment, and health outcomes. Prior work has shown this disconnect between economic and gendered progress is reproduced even within countries: the correlation between per capita income and many of these measures are weak and inconsistent in sign across Indian states. The differential persistence in these gendered norms and their relation with general economic development demand explanation.

In confronting this puzzle, we draw particular inspiration from a nascent political economy literature which uses empirical methods from modern quantitative social science to test long-standing theories of cultural transmission in evolutionary anthropology. Relevant work in this area, for example, has documented that matches in the marriage market can be either hastened or delayed by economic circumstance depending on the direction of marriage payments as determined by prevailing cultural norms (Corno, Hildebrandt, and Voena, ECMA 2020). Operating through a consumption smoothing mechanism that imposes differences in marginal propensities to consume, income shocks are found to incentivize the recipients of marriage payments—the families of grooms in societies which practice dowry or those of brides where bride payments are the custom—to marry earlier. The gendered welfare implications are most salient when considering the resulting effect on the risk of exploitative child marriage. Interestingly, this implies that positive income shocks actually increase the incidence of these arrangements in dowry societies which may contribute to the region's gender regressiveness amidst economic progress.

3.1 Preliminary results

Our first set of results build on this work to study the hazard into marital violence using data from the Demographic and Health Surveys. Figure 3 and Table 1 summarize findings from Indian data. Here, we find that the incidence of a drought perhaps counterintuitively reduces the probability of first-time domestic violence by 6.8-7.4% in the year of the drought's occurrence. Intuitively, the more severe the definition of drought, the larger is the estimated effect. Ongoing work entails exploration of the underlying mechanisms for this effect and the degree to which this effect generalizes across other countries and over time. We are also interested in exploring the effect of environmental shocks on the quality of eventual marriage matches as measured by age gaps, political disagreement, intrahousehold bargaining power, and differences in gendered attitudes.

We also intend to exploit the cohort variation in DHS survey respondents to investigate whether these economically determined deviations from norms hold over time, even generationally transmitted, or transient. To investigate whether the cultural effects of economic shocks are distinctly gendered in nature, we have data on non-gendered (i.e., caste-based and religion-based) attitudes to use as placebo tests. To ascertain whether these effects can partially explain the observed cross-national differences, we are in the process of scaling our domestic violence analysis to the more than 60 other developing countries in the DHS database.

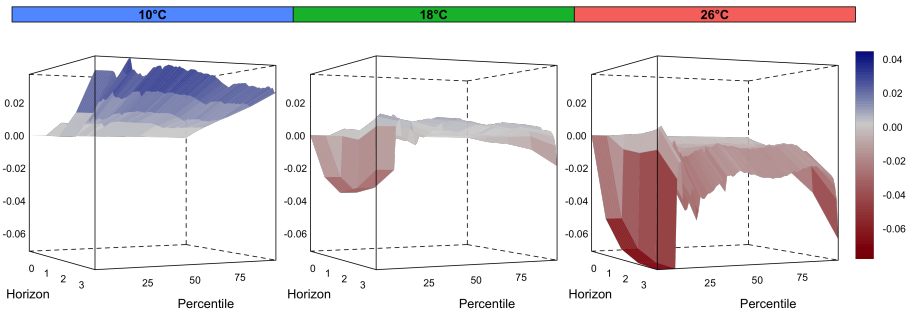
3.2 Future work

In another branch of this project, we explore the formation of various categories of gendered norms, attempting to pin down which norms are most susceptible to change and the most formative periods of life. the timing and the malleability of the formation of gendered norms of other varieties. Another a range of gendered attitudes spanning relationship controlling behavior, justifications for domestic violence, and attitudes towards gendered roles, we seek to investigate what categories of norms are most 'disruptible' and among these, when in a person's lifetime these norms can be most readily disrupted.

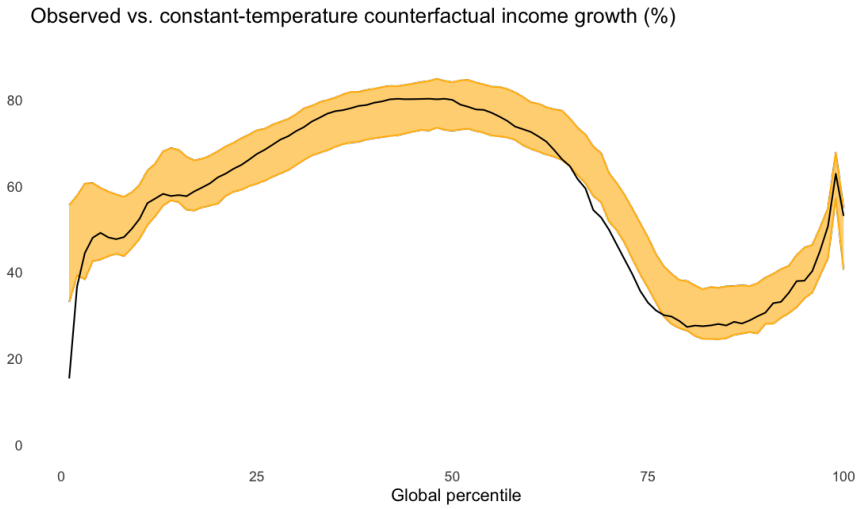
We will also consider cultural disruption on the scale of multiple lifetimes, specifically by measuring the continuation or anti-persistence of these norms from one generation to another and identifying their determinants. We take some inspiration from existing work by Acemoglu et al. (REStud 2024) looking at within-country birth-cohort variation in (non-gendered) national attitudes towards democracy. While they are unable to track individuals, households, or even subnational regions, their analysis indirectly performs a less precise version of the comparative analysis we are keen on implementing ourselves. Those authors use years of birth to infer a personal history of economic shocks to match to their expressed attitudes towards democracy. In our setting where we are able to elicit gendered attitudes from multiple generations of the same household, we can likewise infer their personal history of economic shocks matched with their gendered attitudes and effectively measure the transmission of parents' gendered attitudes as a function of personal experiences with economic shocks essentially racehorsing attitudinal heritability against experiential learning.

Figure 1: Climate change and global income inequality

(a) Cumulative within-country income distributional response to a 1°C shock for different climates



(b) Observed vs. counterfactual global incidence of growth, 1981-2016



(c) Observed vs. counterfactual global inequality, 1981-2016

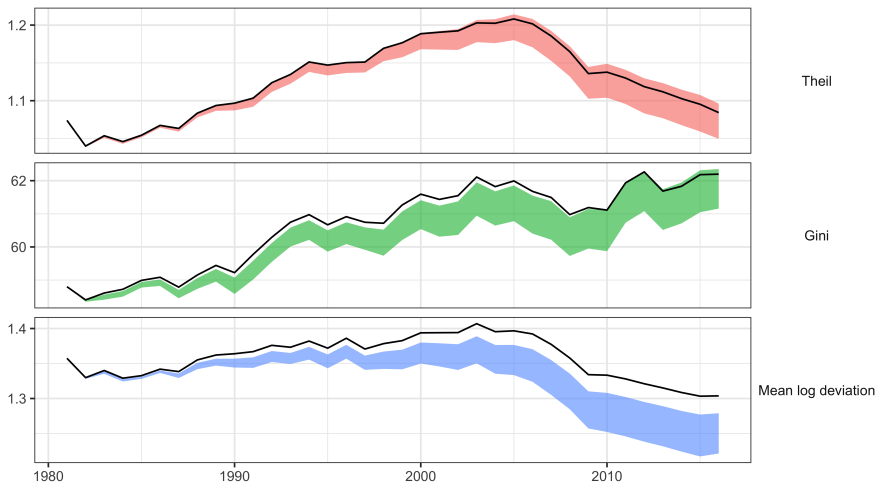
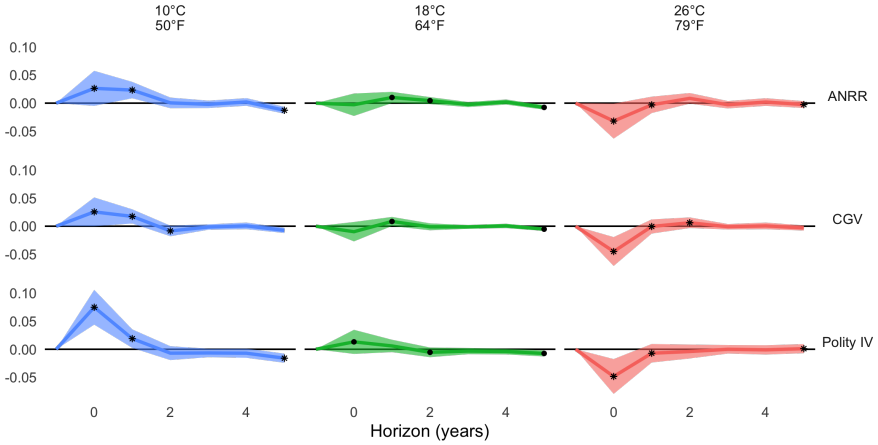


Figure 2: Effect of a 1°C shock on political transition probabilities

(a) Change in the probability of democratic transition among autocracies



(b) Change in the probability of democratic reversal among democracies

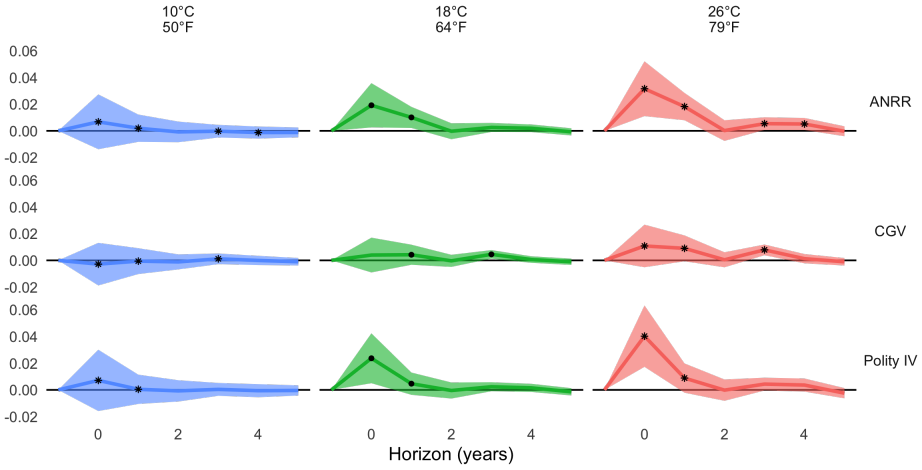


Figure 3: Incidence of domestic violence in India

(a) Baseline annual hazard into first-time marital violence



(b) Effect of different definitions of drought on annual hazard

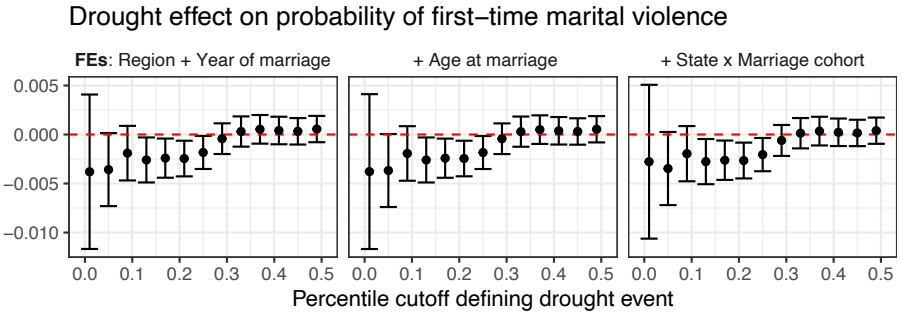


Table 1: Effect of droughts on annual hazard

Dependent Variable:	First violent event	
Drought	-0.002739***	-0.002978***
	(0.000979)	(0.000987)
Year of marriage	Yes	Yes
Age at marriage	Yes	Yes
Respondent age	Yes	Yes
Region	Yes	Yes
State × Marriage cohort		Yes
Observations	355,750	355,750
R^2	0.04374	0.04478
Within R^2	2.13×10^{-5}	2.47×10^{-5}

Signif. Codes: ***: 0.01, **: 0.05, *: 0.1 (Region clustered SEs)